

The Abstraction Gap Between Mathematics and Physics: The Hidden Compromise in Dimensional Analysis Resolved

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0.1 The circular sector

0.1.1 Arc length

$$s = \lim_{n \rightarrow \infty} 2^n s_n \quad (1a)$$

$$s_n = \sqrt{r^2 + r^2 - 2ra_n} \quad (1b)$$

$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}} \quad (1c)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (1d)$$

$$s = \lim_{n \rightarrow \infty} 2^n s_n \quad (2a)$$

$$s_n = \sqrt{2r^2 - 2ra_n} \quad (2b)$$

$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}} \quad (2c)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (2d)$$

$$s = \lim_{n \rightarrow \infty} 2^n s_n \quad (3a)$$

$$s_n = \sqrt{2r^2 - 2ra_n} \quad (3b)$$

$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}} \quad (3c)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (3d)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (4a)$$

$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}} \quad (4b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (4c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (5a)$$

$$a_n = r \frac{\sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}} \quad (5b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (5c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (6a)$$

$$a_n = r \sqrt{\frac{r^2 - a_{n-1}^2}{2r^2 - 2ra_{n-1}}} \quad (6b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (6c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (7a)$$

$$a_n = r \sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r^2 - 2ra_{n-1}}} \quad (7b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (7c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (8a)$$

$$a_n = r \sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r(r - a_{n-1})}} \quad (8b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (8c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (9a)$$

$$a_n = r \sqrt{\frac{r + a_{n-1}}{2r}} \quad (9b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (9c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (10a)$$

$$a_n = r \sqrt{\frac{2r(r + a_{n-1})}{(2r)^2}} \quad (10b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (10c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (11a)$$

$$a_n = r \sqrt{\frac{2r^2 + 2ra_{n-1}}{(2r)^2}} \quad (11b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (11c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (12a)$$

$$a_n = r \sqrt{\frac{\left(\sqrt{2r^2 + 2ra_{n-1}}\right)^2}{(2r)^2}} \quad (12b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (12c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (13a)$$

$$a_n = r \sqrt{\left(\frac{\sqrt{2r^2 + 2ra_{n-1}}}{2r}\right)^2} \quad (13b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (13c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (14a)$$

$$a_n = r \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2r} \quad (14b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (14c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (15a)$$

$$a_n = \frac{r \sqrt{2r^2 + 2ra_{n-1}}}{2r} \quad (15b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (15c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (16a)$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2} \quad (16b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (16c)$$

$$A_{ORS_{n-1}} = \frac{\overline{RS_{n-1}OX_n}}{2} = \frac{\overline{OS_{n-1}RX_{n-1}}}{2} \quad (17)$$

$$\frac{\overline{RS_{n-1}OX_n}}{2} = \frac{\overline{OS_{n-1}RX_{n-1}}}{2} \quad (18)$$

$$\overline{RS_{n-1}OX_n} = \overline{OS_{n-1}RX_{n-1}} \quad (19)$$

$$\frac{\overline{RS_{n-1}OX_n}}{\overline{RS_{n-1}}} = \frac{\overline{OS_{n-1}RX_{n-1}}}{\overline{RS_{n-1}}} \quad (20)$$

$$\overline{OX_n} = \frac{\overline{OS_{n-1}RX_{n-1}}}{\overline{RS_{n-1}}} \quad (21)$$

$$\overline{OX_n} = \frac{\overline{OS_{n-1}} \sqrt{\overline{OR}^2 - \overline{OX_{n-1}}^2}}{\overline{RS_{n-1}}} \quad (22)$$

$$\overline{OX_n} = \frac{\overline{OS_{n-1}} \sqrt{\overline{OR}^2 - \overline{OX_{n-1}}^2}}{\sqrt{\overline{OR}^2 + \overline{OS_{n-1}}^2 - 2\overline{OS_{n-1}OX_{n-1}}}} \quad (23)$$

$$\overline{RS_n}^2 = \overline{RX_n}^2 + \overline{X_nS_n}^2 \quad (24)$$

$$\overline{RS_n} = \sqrt{\overline{RX_n}^2 + \overline{X_nS_n}^2} \quad (25)$$

$$\overline{RS_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + \overline{X_nS_n}^2} \quad (26)$$

$$\overline{RS_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + (\overline{OS_n} - \overline{OX_n})^2} \quad (27)$$

$$\overline{RS_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + \overline{OS_n}^2 - 2\overline{OS_nOX_n} + \overline{OX_n}^2} \quad (28)$$

$$\overline{RS_n} = \sqrt{\overline{OR}^2 + \overline{OS_n}^2 - 2\overline{OS_nOX_n}} \quad (29)$$

$$\overline{OX_n}^2 + \overline{RX_n}^2 = \overline{OR}^2 \quad (30)$$

$$\overline{OX_n}^2 + \overline{RX_n}^2 - \overline{OX_n}^2 = \overline{OR}^2 - \overline{OX_n}^2 \quad (31)$$

$$\overline{RX_n}^2 = \overline{OR}^2 - \overline{OX_n}^2 \quad (32)$$

$$\overline{RX_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2} \quad (33)$$

0.1.2 Area

$$A = \frac{rs}{2} \quad (34)$$

$$A_0 = \lim_{n \rightarrow \infty} 2^{n-1} A_{ORS_nS_{n-1}} \quad (35)$$

$$A_0 = \lim_{n \rightarrow \infty} 2^{n-1} \frac{\overline{RS_{n-1}OS_n}}{2} \quad (36)$$

$$A_0 = \frac{\lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1}OS_n}}{2} \quad (37)$$

$$A_0 = \frac{\lim_{n \rightarrow \infty} (2^{n-1} \overline{RS_{n-1}}) \overline{OS_n}}{2} \quad (38)$$

$$A_0 = \frac{\lim_{n \rightarrow \infty} \overline{OS_n} (2^{n-1} \overline{RS_{n-1}})}{2} \quad (39)$$

$$A_0 = \frac{\left(\lim_{n \rightarrow \infty} \overline{OS_n} \right) \left(\lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1}} \right)}{2} \quad (40)$$

$$A_0 = \frac{\overline{OR} \lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1}}}{2} \quad (41)$$

$$A_0 = \frac{\overline{OR} \text{arc } RS_0}{2} \quad (42)$$

0.1.3 The circular functions

0.1.3.1 The sine function

$$\begin{aligned} \sin: \{(r, s) \in \mathbb{R}^2 \mid r > 0\} &\rightarrow \mathbb{R} \\ (r, s) &\mapsto y \end{aligned} \quad (43)$$

$$\cos(r, \alpha - \beta) = a_{0,0} \quad (44a)$$

$$a_{0,0} = r - \frac{s^2}{2^{2 \cdot 0 + 1} r} \quad (44b)$$

$$s = \sqrt{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2} \quad (44c)$$

$$\cos(r, \alpha - \beta) = a_{0,0} \quad (45a)$$

$$a_{0,0} = r - \frac{s^2}{2r} \quad (45b)$$

$$s = \sqrt{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2} \quad (45c)$$

$$\cos(r, \alpha - \beta) = a_{0,0} \quad (46a)$$

$$a_{0,0} = r - \frac{\left(\sqrt{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2} \right)^2}{2r} \quad (46b)$$

$$\cos(r, \alpha - \beta) = r - \frac{\left(\sqrt{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2} \right)^2}{2r} \quad (47)$$

$$\cos(r, \alpha - \beta) = r - \frac{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2}{2r} \quad (48)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2}{2r} - \frac{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2}{2r} \quad (49)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - ((\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2)}{2r} \quad (50)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos(r, \alpha) - \cos(r, \beta))^2 - (\sin(r, \alpha) - \sin(r, \beta))^2}{2r} \quad (51)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos^2(r, \alpha) - 2\cos(r, \alpha)\cos(r, \beta) + \cos^2(r, \beta)) - (\sin(r, \alpha) - \sin(r, \beta))^2}{2r} \quad (52)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos^2(r, \alpha) - 2\cos(r, \alpha)\cos(r, \beta) + \cos^2(r, \beta)) - (\sin^2(r, \alpha) - 2\sin(r, \alpha)\sin(r, \beta) + \sin^2(r, \beta))}{2r} \quad (53)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - \cos^2(r, \alpha) + 2\cos(r, \alpha)\cos(r, \beta) - \cos^2(r, \beta) - (\sin^2(r, \alpha) - 2\sin(r, \alpha)\sin(r, \beta) + \sin^2(r, \beta))}{2r} \quad (54)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - \cos^2(r, \alpha) + 2\cos(r, \alpha)\cos(r, \beta) - \cos^2(r, \beta) - \sin^2(r, \alpha) + 2\sin(r, \alpha)\sin(r, \beta) - \sin^2(r, \beta)}{2r} \quad (55)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos^2(r, \alpha) + \sin^2(r, \alpha)) + 2\cos(r, \alpha)\cos(r, \beta) - \cos^2(r, \beta) + 2\sin(r, \alpha)\sin(r, \beta) - \sin^2(r, \beta)}{2r} \quad (56)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos^2(r, \alpha) + \sin^2(r, \alpha)) - (\cos^2(r, \beta) + \sin^2(r, \beta)) + 2\cos(r, \alpha)\cos(r, \beta) + 2\sin(r, \alpha)\sin(r, \beta)}{2r} \quad (57)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos^2(r, \alpha) + \sin^2(r, \alpha)) - (\cos^2(r, \beta) + \sin^2(r, \beta)) + 2(\cos(r, \alpha)\cos(r, \beta) + \sin(r, \alpha)\sin(r, \beta))}{2r} \quad (58)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - r^2 - (\cos^2(r, \beta) + \sin^2(r, \beta)) + 2(\cos(r, \alpha)\cos(r, \beta) + \sin(r, \alpha)\sin(r, \beta))}{2r} \quad (59)$$

$$\cos(r, \alpha - \beta) = \frac{r^2 - (\cos^2(r, \beta) + \sin^2(r, \beta)) + 2(\cos(r, \alpha)\cos(r, \beta) + \sin(r, \alpha)\sin(r, \beta))}{2r} \quad (60)$$

$$\cos(r, \alpha - \beta) = \frac{r^2 - r^2 + 2(\cos(r, \alpha)\cos(r, \beta) + \sin(r, \alpha)\sin(r, \beta))}{2r} \quad (61)$$

$$\cos(r, \alpha - \beta) = \frac{2(\cos(r, \alpha)\cos(r, \beta) + \sin(r, \alpha)\sin(r, \beta))}{2r} \quad (62)$$

$$\cos(r, \alpha - \beta) = \frac{\cos(r, \alpha)\cos(r, \beta) + \sin(r, \alpha)\sin(r, \beta)}{r} \quad (63)$$

$$\cos(r, \alpha - \beta) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha - \beta}{r}\right)^{2k}}{(2k)!} \quad (64)$$

$$\cos(r, \alpha - \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \left(\frac{\alpha - \beta}{r}\right)^{2k}}{(2k)!} \quad (65)$$

$$\cos(r, \alpha - \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k} \binom{2k}{j} \left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^j}{(2k)!} \quad (66)$$

$$\cos(r, \alpha - \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k} \frac{(2k)!}{j!(2k-j)!} \left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^j}{(2k)!} \quad (67)$$

$$\cos(r, \alpha - \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k} (2k)! \frac{\left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^j}{j!(2k-j)!}}{(2k)!} \quad (68)$$

$$\cos(r, \alpha - \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k (2k)! \sum_{j=0}^{2k} \frac{\left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^j}{j!(2k-j)!}}{(2k)!} \quad (69)$$

$$\cos(r, \alpha - \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n (-1)^k \sum_{j=0}^{2k} \frac{\left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^j}{j!(2k-j)!} \quad (70)$$

$$\cos(r, \alpha - \beta) = r \lim_{n \rightarrow \infty} \left(\sum_{k=0}^n \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{j=0}^{n-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j}}{(2j)!} + \sum_{k=0}^{n-1} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-1-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!} \right) \quad (71)$$

$$\cos(r, \alpha - \beta) = r \left(\lim_{n \rightarrow \infty} \left(\sum_{k=0}^n \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{j=0}^{n-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j}}{(2j)!} \right) + \lim_{n \rightarrow \infty} \left(\sum_{k=0}^{n-1} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-1-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!} \right) \right) \quad (72)$$

$$\cos(r, \alpha - \beta) = r \left(\sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} + \lim_{n \rightarrow \infty} \left(\sum_{k=0}^{n-1} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-1-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!} \right) \right) \quad (73)$$

$$\cos(r, \alpha - \beta) = r \left(\sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \right) \quad (74)$$

$$\cos(r, \alpha - \beta) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} + r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \quad (75)$$

$$\cos(r, \alpha - \beta) = \frac{r \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} + r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \right)}{r} \quad (76)$$

$$\cos(r, \alpha - \beta) = \frac{r^2 \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} + r^2 \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}}{r} \quad (77)$$

$$\cos(r, \alpha - \beta) = \frac{\left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} \right) + r^2 \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}}{r} \quad (78)$$

$$\cos(r, \alpha - \beta) = \frac{\left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} \right) + \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \right)}{r} \quad (79)$$

$$\cos(r, \alpha - \beta) = \frac{\cos(r, \alpha) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} \right) + \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \right)}{r} \quad (80)$$

$$\cos(r, \alpha - \beta) = \frac{\cos(r, \alpha) \cos(r, \beta) + \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \right)}{r} \quad (81)$$

$$\sin(r, \alpha) \sin(r, \beta) = \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \right) \quad (82)$$

$$\begin{cases} \sin(r, \alpha) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \\ \sin(r, \beta) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \end{cases} \quad (83)$$

$$\sin(r, s) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \quad (84)$$

Properties

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \right)}{\partial s} \quad (85)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k+1}}{r^{2k+1}}}{(2k+1)!} \right)}{\partial s} \quad (86)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k+1} (2k+1)!} \right)}{\partial s} \quad (87)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k r s^{2k+1}}{r^{2k+1} (2k+1)!} \right)}{\partial s} \quad (88)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k} (2k+1)!} \right)}{\partial s} \quad (89)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k+1}}{\partial s}}{r^{2k} (2k+1)!} \quad (90)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) s^{2k}}{r^{2k} (2k+1)!} \quad (91)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \frac{s^{2k}}{r^{2k}}}{(2k+1)!} \quad (92)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r}\right)^{2k}}{(2k+1)!} \quad (93)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r}\right)^{2k}}{(2k+1)(2k)!} \quad (94)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \quad (95)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}}{r} \quad (96)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\cos(r, s)}{r} \quad (97)$$

0.1.3.2 The arcsine function

$$\begin{aligned} \arcsin: \{(r, y) \in \mathbb{R}^2 \mid r > 0, -r \leq y \leq r\} &\rightarrow \mathbb{R} \\ (r, y) &\mapsto s \end{aligned} \quad (98)$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) + \arcsin(r, y)}{2} \quad (99)$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) + \arcsin(r, y) + \arccos(r, y) - \arccos(r, y)}{2} \quad (100)$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) + \arcsin(r, -y) + \arccos(r, -y) - \arccos(r, y)}{2} \quad (101)$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) - \arcsin(r, y) + \arccos(r, -y) - \arccos(r, y)}{2} \quad (102)$$

$$\arcsin(r, y) = \frac{\arccos(r, -y) - \arccos(r, y)}{2} \quad (103)$$

0.1.3.3 The cosine function

$$\begin{aligned} \cos: \{(r, s) \in \mathbb{R}^2 \mid r > 0\} &\rightarrow \mathbb{R} \\ (r, s) &\mapsto x \end{aligned} \quad (104)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (105a)$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2} \quad (105b)$$

$$a_0 = \cos(r, s) \quad (105c)$$

$$s = \lim_{m \rightarrow \infty} 2^m \sqrt{2r^2 - 2ra_{m,m}} \quad (106a)$$

$$a_{m,n} = \frac{\sqrt{2r^2 + 2ra_{m,n-1}}}{2} \quad \text{for } n \leq m \quad (106b)$$

$$a_{m,0} = \cos(r, s) \quad (106c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (107a)$$

$$a_{m,n-1} = \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2} \quad \text{for } n \leq m \quad (107b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (107c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (108a)$$

$$2a_{m,n-1} = 2 \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2} \quad \text{for } n \leq m \quad (108b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (108c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (109a)$$

$$2a_{m,n-1} = \frac{2\sqrt{2r^2 + 2ra_{m,n}}}{2} \quad \text{for } n \leq m \quad (109b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (109c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (110a)$$

$$2a_{m,n-1} = \sqrt{2r^2 + 2ra_{m,n}} \quad \text{for } n \leq m \quad (110b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (110c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (111a)$$

$$(2a_{m,n-1})^2 = \left(\sqrt{2r^2 + 2ra_{m,n}} \right)^2 \quad \text{for } n \leq m \quad (111b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (111c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (112a)$$

$$2^2 a_{m,n-1}^2 = \left(\sqrt{2r^2 + 2ra_{m,n}} \right)^2 \quad \text{for } n \leq m \quad (112b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (112c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (113a)$$

$$4a_{m,n-1}^2 = \left(\sqrt{2r^2 + 2ra_{m,n}} \right)^2 \quad \text{for } n \leq m \quad (113b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (113c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (114a)$$

$$4a_{m,n-1}^2 = 2r^2 + 2ra_{m,n} \quad \text{for } n \leq m \quad (114b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (114c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (115a)$$

$$4a_{m,n-1}^2 - 2r^2 = 2r^2 + 2ra_{m,n} - 2r^2 \quad \text{for } n \leq m \quad (115b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (115c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (116a)$$

$$4a_{m,n-1}^2 - 2r^2 = 2ra_{m,n} \quad \text{for } n \leq m \quad (116b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (116c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (117a)$$

$$\frac{4a_{m,n-1}^2 - 2r^2}{2r} = \frac{2ra_{m,n}}{2r} \quad \text{for } n \leq m \quad (117b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (117c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (118a)$$

$$\frac{4a_{m,n-1}^2}{2r} - \frac{2r^2}{2r} = \frac{2ra_{m,n}}{2r} \quad \text{for } n \leq m \quad (118b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (118c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (119a)$$

$$\frac{2a_{m,n-1}^2}{r} - \frac{2r^2}{2r} = \frac{2ra_{m,n}}{2r} \quad \text{for } n \leq m \quad (119b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (119c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (120a)$$

$$\frac{2a_{m,n-1}^2}{r} - r = \frac{2ra_{m,n}}{2r} \quad \text{for } n \leq m \quad (120b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (120c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (121a)$$

$$\frac{2a_{m,n-1}^2}{r} - r = a_{m,n} \quad \text{for } n \leq m \quad (121b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (121c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (122a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (122b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (122c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (123a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (123b)$$

$$\frac{2^m \sqrt{2r^2 - 2ra_{m,0}}}{2^m} = \frac{s}{2^m} \quad (123c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (124a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (124b)$$

$$\sqrt{2r^2 - 2ra_{m,0}} = \frac{s}{2^m} \quad (124c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (125a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (125b)$$

$$\left(\sqrt{2r^2 - 2ra_{m,0}} \right)^2 = \left(\frac{s}{2^m} \right)^2 \quad (125c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (126a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (126b)$$

$$2r^2 - 2ra_{m,0} = \left(\frac{s}{2^m}\right)^2 \quad (126c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (127a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (127b)$$

$$2r^2 - 2ra_{m,0} = \frac{s^2}{(2^m)^2} \quad (127c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (128a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (128b)$$

$$2r^2 - 2ra_{m,0} = \frac{s^2}{2^{2m}} \quad (128c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (129a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (129b)$$

$$2r^2 - 2ra_{m,0} - 2r^2 = \frac{s^2}{2^{2m}} - 2r^2 \quad (129c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (130a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (130b)$$

$$-2ra_{m,0} = \frac{s^2}{2^{2m}} - 2r^2 \quad (130c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (131a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (131b)$$

$$\frac{-2ra_{m,0}}{-2r} = \frac{\left(\frac{s^2}{2^{2m}}\right) - 2r^2}{-2r} \quad (131c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (132a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (132b)$$

$$a_{m,0} = \frac{\left(\frac{s^2}{2^{2m}}\right) - 2r^2}{-2r} \quad (132c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (133a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (133b)$$

$$a_{m,0} = \frac{\left(\frac{s^2}{2^{2m}}\right)}{-2r} - \frac{2r^2}{-2r} \quad (133c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (134a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (134b)$$

$$a_{m,0} = \frac{s^2}{2^{2m}(-2r)} - \frac{2r^2}{-2r} \quad (134c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (135a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (135b)$$

$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} - \frac{2r^2}{-2r} \quad (135c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (136a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (136b)$$

$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} + \frac{-2r^2}{-2r} \quad (136c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (137a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (137b)$$

$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} + r \quad (137c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (138a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (138b)$$

$$a_{m,0} = r + \frac{s^2}{-2^{2m+1}r} \quad (138c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (139a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (139b)$$

$$a_{m,0} = r - \left(-\frac{s^2}{-2^{2m+1}r} \right) \quad (139c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (140a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (140b)$$

$$a_{m,0} = r - \left(\frac{-s^2}{-2^{2m+1}r} \right) \quad (140c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (141a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (141b)$$

$$a_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (141c)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (142a)$$

$$a_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (142b)$$

$$r\hat{a}_{m,n} = \frac{2(r\hat{a}_{m,n-1})^2}{r} - r \quad \text{for } n \leq m \quad (143a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (143b)$$

$$r\hat{a}_{m,n} = \frac{2r^2\hat{a}_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (144a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (144b)$$

$$r\hat{a}_{m,n} = 2r\hat{a}_{m,n-1}^2 - r \quad \text{for } n \leq m \quad (145a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (145b)$$

$$\frac{r\hat{a}_{m,n}}{r} = \frac{2r\hat{a}_{m,n-1}^2 - r}{r} \quad \text{for } n \leq m \quad (146a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (146b)$$

$$\hat{a}_{m,n} = \frac{2r\hat{a}_{m,n-1}^2 - r}{r} \quad \text{for } n \leq m \quad (147a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (147b)$$

$$\hat{a}_{m,n} = \frac{2r\hat{a}_{m,n-1}^2}{r} - \frac{r}{r} \quad \text{for } n \leq m \quad (148a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (148b)$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - \frac{r}{r} \quad \text{for } n \leq m \quad (149a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (149b)$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m \quad (150a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (150b)$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m \quad (151a)$$

$$\frac{r\hat{a}_{m,0}}{r} = \frac{r - \left(\frac{s^2}{2^{2m+1}r}\right)}{r} \quad (151b)$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m \quad (152a)$$

$$\hat{a}_{m,0} = \frac{r - \left(\frac{s^2}{2^{2m+1}r}\right)}{r} \quad (152b)$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m \quad (153a)$$

$$\hat{a}_{m,0} = \frac{r}{r} - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r} \quad (153b)$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m \quad (154a)$$

$$\hat{a}_{m,0} = 1 - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r} \quad (154b)$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m \quad (155a)$$

$$\hat{a}_{m,0} = 1 - \frac{s^2}{2^{2m+1}r^2} \quad (155b)$$

$$\hat{a}_{m,n} = T_2^{\circ n}(\hat{a}_{m,0}) \quad \text{for } n \leq m \quad (156)$$

$$\hat{a}_{m,n} = T_2^{\circ n} \left(1 - \frac{s^2}{2^{2m+1}r^2} \right) \quad \text{for } n \leq m \quad (157)$$

$$\hat{a}_{m,n} = T_{2^n} \left(1 - \frac{s^2}{2^{2m+1}r^2} \right) \quad \text{for } n \leq m \quad (158)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(\left(1 - \frac{s^2}{2^{2m+1}r^2} \right) - 1 \right)^k \quad \text{for } n \leq m \quad (159)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(1 - \frac{s^2}{2^{2m+1}r^2} - 1 \right)^k \quad \text{for } n \leq m \quad (160)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(-\frac{s^2}{2^{2m+1}r^2} \right)^k \quad \text{for } n \leq m \quad (161)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \left(-\frac{s^2}{2^{2m+1}r^2} \right)^k \quad \text{for } n \leq m \quad (162)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \left(-\frac{s^2}{2^{2m+1}r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m \quad (163)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(2 \left(-\frac{s^2}{2^{2m+1}r^2} \right) \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m \quad (164)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-2 \frac{s^2}{2^{2m+1}r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m \quad (165)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-\frac{2s^2}{2^{2m+1}r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m \quad (166)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{2^{2m}r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m \quad (167)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{(2^2)^m r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m \quad (168)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{4^m r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m \quad (169)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s^2}{4^m r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m \quad (170)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{\left(\frac{s}{r} \right)^2}{4^m} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m \quad (171)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{\left(\frac{s}{r} \right)^2}{4^m} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m \quad (172)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \frac{\left(\frac{s}{r}\right)^{2k}}{(4^m)^k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m \quad (173)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \frac{\left(\frac{s}{r}\right)^{2k}}{(4^m)^k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m \quad (174)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(4^m)^k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m \quad (175)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(4^m)^k} \right)}{(2k)!} \quad \text{for } n \leq m \quad (176)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(4^m)^k (2k)!} \quad \text{for } n \leq m \quad (177)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)! (4^m)^k} \quad \text{for } n \leq m \quad (178)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)! (4^m)^k} \quad \text{for } n \leq m \quad (179)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)! \prod_{j=0}^{k-1} 4^m} \quad \text{for } n \leq m \quad (180)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{(2^n)^2 - j^2}{4^m} \quad \text{for } n \leq m \quad (181)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{(2^n)^2 - j^2}{4^m} \quad \text{for } n \leq m \quad (182)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{4^n - j^2}{4^m} \quad \text{for } n \leq m \quad (183)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(\frac{4^n}{4^m} - \frac{j^2}{4^m} \right) \quad \text{for } n \leq m \quad (184)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m} \right) \quad \text{for } n \leq m \quad (185)$$

$$r\hat{a}_{m,n} = r \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m}\right) \quad \text{for } n \leq m \quad (186)$$

$$a_{m,n} = r \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m}\right) \quad \text{for } n \leq m \quad (187)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (188)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{m-m} - \frac{j^2}{4^m}\right) \quad (189)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^0 - \frac{j^2}{4^m}\right) \quad (190)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^m}\right) \quad (191)$$

$$\cos(r, s) = r \lim_{m \rightarrow \infty} \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^m}\right) \quad (192)$$

$$\cos(r, s) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \quad (193)$$

Properties

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \right)}{\partial s} \quad (194)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k}}{r^{2k}}}{(2k)!} \right)}{\partial s} \quad (195)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k s^{2k}}{r^{2k} (2k)!} \right)}{\partial s} \quad (196)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k r s^{2k}}{r^{2k} (2k)!} \right)}{\partial s} \quad (197)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k s^{2k}}{r^{2k-1} (2k)!} \right)}{\partial s} \quad (198)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k}}{\partial s}}{r^{2k-1} (2k)!} \quad (199)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) s^{2k-1}}{r^{2k-1} (2k)!} \quad (200)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \frac{s^{2k-1}}{r^{2k-1}}}{(2k)!} \quad (201)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \left(\frac{s}{r}\right)^{2k-1}}{(2k)!} \quad (202)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \left(\frac{s}{r}\right)^{2k-1}}{(2k)(2k-1)!} \quad (203)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \quad (204)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=1}^{\infty} \frac{-(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \quad (205)$$

$$\frac{\partial \cos(r, s)}{\partial s} = - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \quad (206)$$

$$\frac{\partial \cos(r, s)}{\partial s} = - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2k-1)!} \quad (207)$$

$$\frac{\partial \cos(r, s)}{\partial s} = - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2(k-1)+1)!} \quad (208)$$

$$\frac{\partial \cos(r, s)}{\partial s} = - \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \quad (209)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{-r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!}}{r} \quad (210)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{-\sin(r, s)}{r} \quad (211)$$

0.1.3.4 The arccosine function

$$\begin{aligned} \arccos: \{(r, x) \in \mathbb{R}^2 \mid r > 0, -r \leq x \leq r\} &\rightarrow \mathbb{R} \\ (r, x) &\mapsto s \end{aligned} \quad (212)$$

$$\arccos(r, x) = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (213a)$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2} \quad (213b)$$

$$a_0 = x \quad (213c)$$

0.2 Simple harmonic motion

$$[m] = \text{kg} \quad (214)$$

$$[\omega] = \text{m/s} \quad (215)$$

$$[k] = \text{N/m} \quad (216)$$

$$[r] = \text{m} \quad (217)$$

$$F = ma = -kx \quad (218)$$

$$ma = -kx \quad (219)$$

$$x(t) = \cos(r, \omega t) \quad (220)$$

$$a(t) = \frac{d^2 x}{dt^2} \quad (221)$$

$$a(t) = \frac{d^2 \cos(r, \omega t)}{dt^2} \quad (222)$$

$$a(t) = \frac{-\omega^2 \cos(r, \omega t)}{r^2} \quad (223)$$

$$a(x) = \frac{-\omega^2 x}{r^2} \quad (224)$$

$$m \frac{-\omega^2 x}{r^2} = -kx \quad (225)$$

$$\frac{m(-\omega^2 x)}{r^2} = -kx \quad (226)$$

$$\frac{-m\omega^2 x}{r^2} = -kx \quad (227)$$

$$\frac{-m\omega^2 x}{r^2} = \frac{-kr^2 x}{r^2} \quad (228)$$

$$m\omega^2 = kr^2 \quad (229)$$

$$v(t) = \frac{dx}{dt} \quad (230)$$

$$v(t) = \frac{d \cos(r, \omega t)}{dt} \quad (231)$$

$$v(t) = \frac{-\omega \sin(r, \omega t)}{r} \quad (232)$$

$$K(t) = \frac{mv^2(t)}{2} \quad (233)$$

$$K(t) = \frac{m \left(\frac{-\omega \sin(r, \omega t)}{r} \right)^2}{2} \quad (234)$$

$$K(t) = \frac{m \frac{(-\omega \sin(r, \omega t))^2}{r^2}}{2} \quad (235)$$

$$K(t) = \frac{m \frac{(-\omega)^2 \sin^2(r, \omega t)}{r^2}}{2} \quad (236)$$

$$K(t) = \frac{m \frac{\omega^2 \sin^2(r, \omega t)}{r^2}}{2} \quad (237)$$

$$K(t) = \frac{\frac{m\omega^2 \sin^2(r, \omega t)}{r^2}}{2} \quad (238)$$

$$K(t) = \frac{\frac{kr^2 \sin^2(r, \omega t)}{r^2}}{2} \quad (239)$$

$$K(t) = \frac{k \sin^2(r, \omega t)}{2} \quad (240)$$

$$U(t) = \frac{kx^2(t)}{2} \quad (241)$$

$$U(t) = \frac{k \cos^2(r, \omega t)}{2} \quad (242)$$

$$E = K + U \quad (243)$$

$$E = \frac{k \sin^2(r, \omega t)}{2} + \frac{k \cos^2(r, \omega t)}{2} \quad (244)$$

$$E = \frac{k \sin^2(r, \omega t) + k \cos^2(r, \omega t)}{2} \quad (245)$$

$$E = \frac{k(\sin^2(r, \omega t) + \cos^2(r, \omega t))}{2} \quad (246)$$

$$E = \frac{kr^2}{2} \quad (247)$$

$$E = \frac{m\omega^2}{2} \quad (248)$$

0.3 Uniform circular motion

$$[m] = \text{kg} \quad (249)$$

$$[\omega] = \text{m/s} \quad (250)$$

$$[r] = \text{m} \quad (251)$$

$$F = ma \quad (252)$$

$$\omega = \frac{\Delta s}{\Delta t} \quad (253)$$

$$\vec{r}_x(t) = \cos(r, \omega t) \quad (254)$$

$$\vec{r}_y(t) = \sin(r, \omega t) \quad (255)$$

$$\vec{r} = (\vec{r}_x, \vec{r}_y) \quad (256)$$

$$\vec{r}(t) = (\cos(r, \omega t), \sin(r, \omega t)) \quad (257)$$

$$\vec{v} = \frac{d\vec{r}}{dt} \quad (258)$$

$$\vec{v}(t) = \frac{d(\cos(r, \omega t), \sin(r, \omega t))}{dt} \quad (259)$$

$$\vec{v}(t) = \left(\frac{d \cos(r, \omega t)}{dt}, \frac{d \sin(r, \omega t)}{dt} \right) \quad (260)$$

$$\vec{v}(t) = \left(\frac{-\omega \sin(r, \omega t)}{r}, \frac{\omega \cos(r, \omega t)}{r} \right) \quad (261)$$

$$\vec{v}(t) = \frac{\omega (-\sin(r, \omega t), \cos(r, \omega t))}{r} \quad (262)$$

$$\vec{a} = \frac{d^2 \vec{r}}{dt^2} \quad (263)$$

$$\vec{a}(t) = \frac{d^2 (\cos(r, \omega t), \sin(r, \omega t))}{dt^2} \quad (264)$$

$$\vec{a}(t) = \left(\frac{d^2 \cos(r, \omega t)}{dt^2}, \frac{d^2 \sin(r, \omega t)}{dt^2} \right) \quad (265)$$

$$\vec{a}(t) = \left(\frac{-\omega^2 \cos(r, \omega t)}{r^2}, \frac{-\omega^2 \sin(r, \omega t)}{r^2} \right) \quad (266)$$

$$\vec{a}(t) = \frac{-\omega^2 (\cos(r, \omega t), \sin(r, \omega t))}{r^2} \quad (267)$$

$$\vec{a} = \frac{-\omega^2 \vec{r}}{r^2} \quad (268)$$

$$\vec{F} = m \vec{a} \quad (269)$$

$$\vec{F} = m \frac{-\omega^2 \vec{r}}{r^2} \quad (270)$$

$$\vec{F} = \frac{m(-\omega^2 \vec{r})}{r^2} \quad (271)$$

$$\vec{F} = \frac{-m\omega^2 \vec{r}}{r^2} \quad (272)$$

$$v = |\vec{v}| \quad (273)$$

$$v = \left| \frac{\omega (-\sin(r, \omega t), \cos(r, \omega t))}{r} \right| \quad (274)$$

$$v = \frac{|\omega (-\sin(r, \omega t), \cos(r, \omega t))|}{|r|} \quad (275)$$

$$v = \frac{|\omega| |(-\sin(r, \omega t), \cos(r, \omega t))|}{|r|} \quad (276)$$

$$v = \frac{\omega |(-\sin(r, \omega t), \cos(r, \omega t))|}{|r|} \quad (277)$$

$$v = \frac{\omega \sqrt{(-\sin(r, \omega t))^2 + \cos^2(r, \omega t)}}{|r|} \quad (278)$$

$$v = \frac{\omega \sqrt{\sin^2(r, \omega t) + \cos^2(r, \omega t)}}{|r|} \quad (279)$$

$$v = \frac{\omega \sqrt{r^2}}{|r|} \quad (280)$$

$$v = \frac{\omega |r|}{|r|} \quad (281)$$

$$v = \omega \quad (282)$$

$$E = \frac{mv^2}{2} \quad (283)$$

$$E = \frac{m\omega^2}{2} \quad (284)$$

$$a = |\vec{a}| \quad (285)$$

$$a = \left| \frac{-\omega^2 \vec{r}}{r^2} \right| \quad (286)$$

$$a = \frac{|-\omega^2 \vec{r}|}{|r^2|} \quad (287)$$

$$a = \frac{|-\omega^2| |\vec{r}|}{|r^2|} \quad (288)$$

$$a = \frac{\omega^2 |\vec{r}|}{|r^2|} \quad (289)$$

$$a = \frac{\omega^2 r}{|r^2|} \quad (290)$$

$$a = \frac{\omega^2 r}{r^2} \quad (291)$$

$$a = \frac{\omega^2}{r} \quad (292)$$

$$F = \frac{m\omega^2}{r} \quad (293)$$

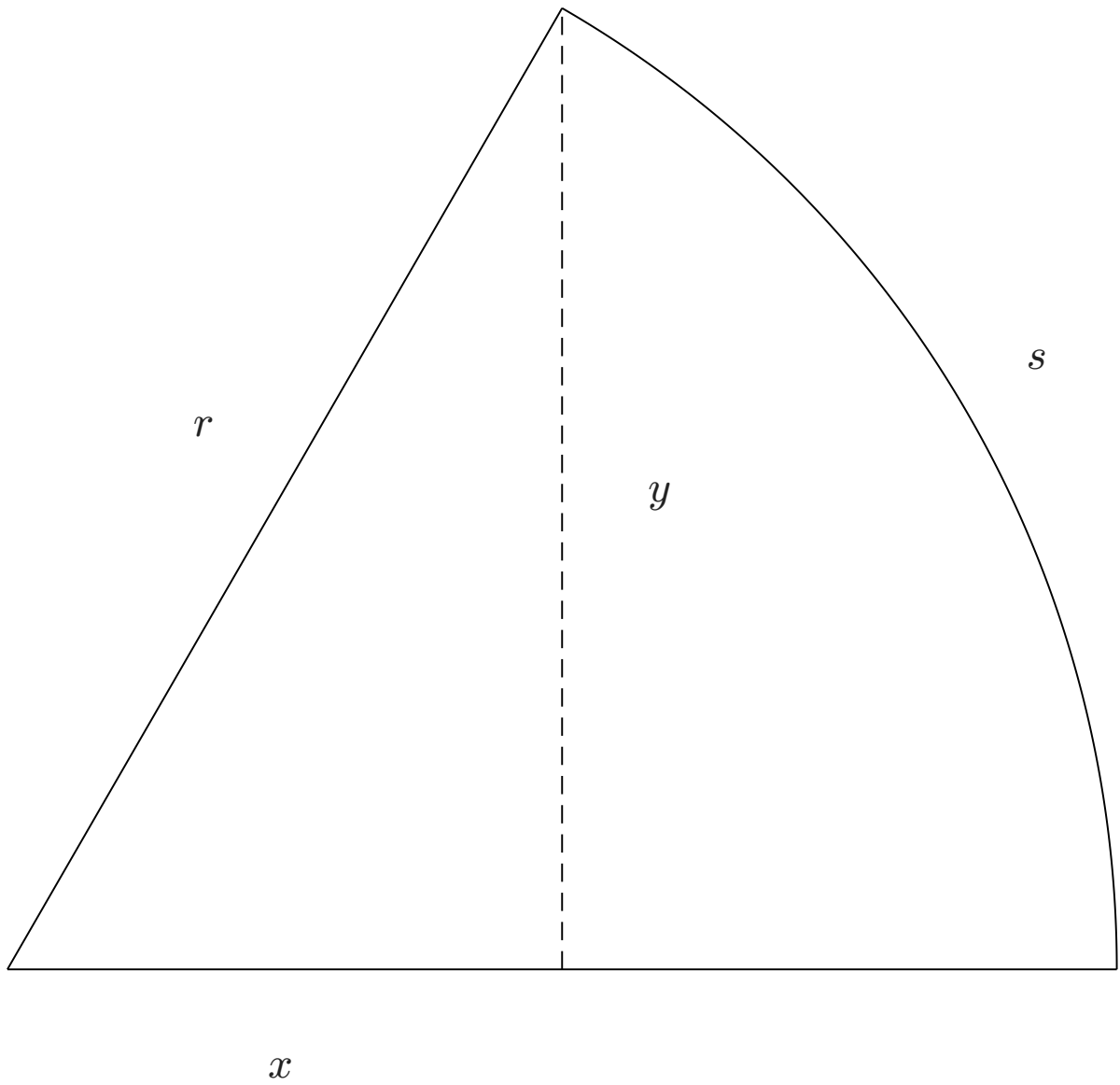
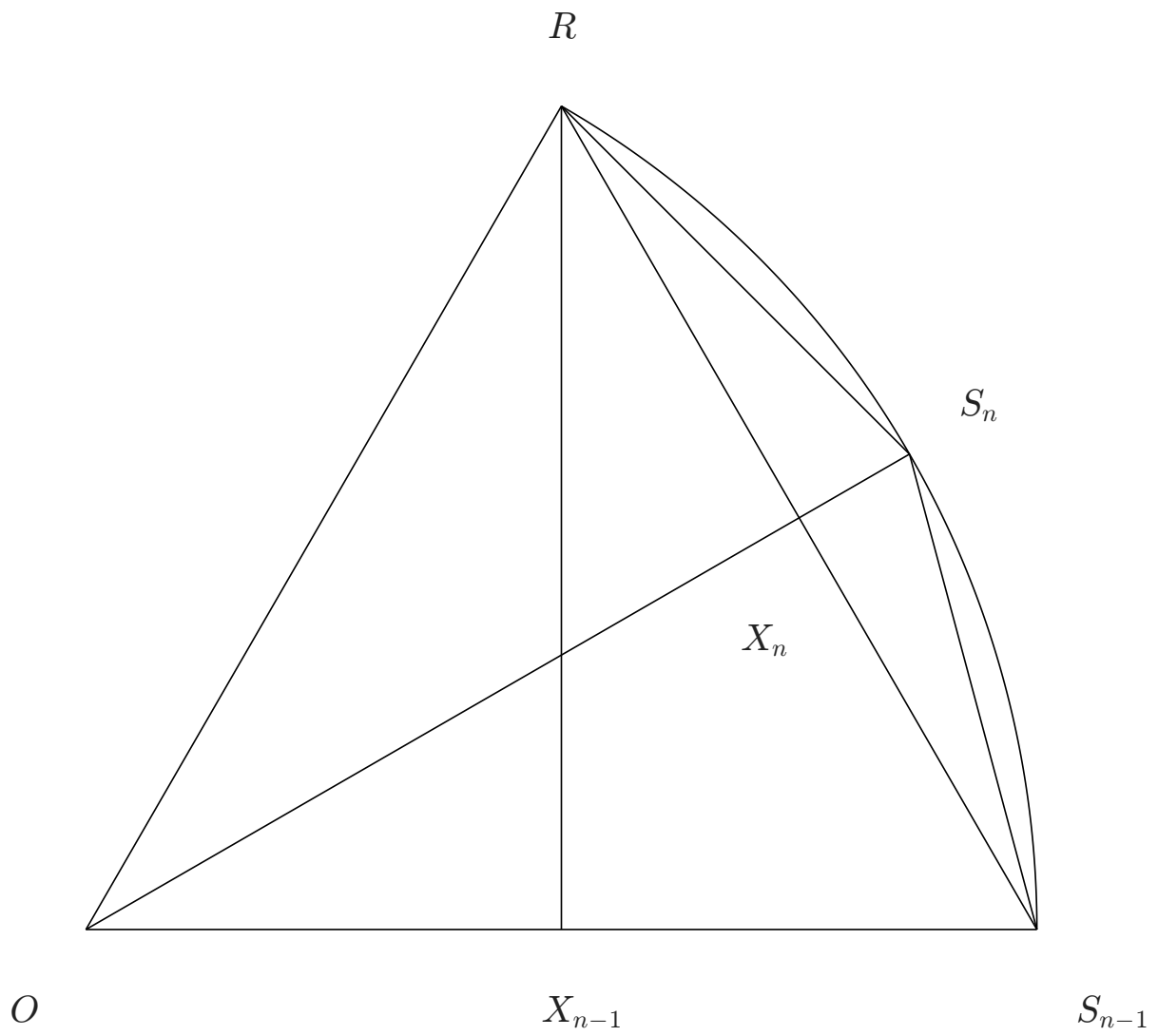


Figure 1: Circular sector.



(a) Recursive step.

Figure 2: Circular sector sequence.

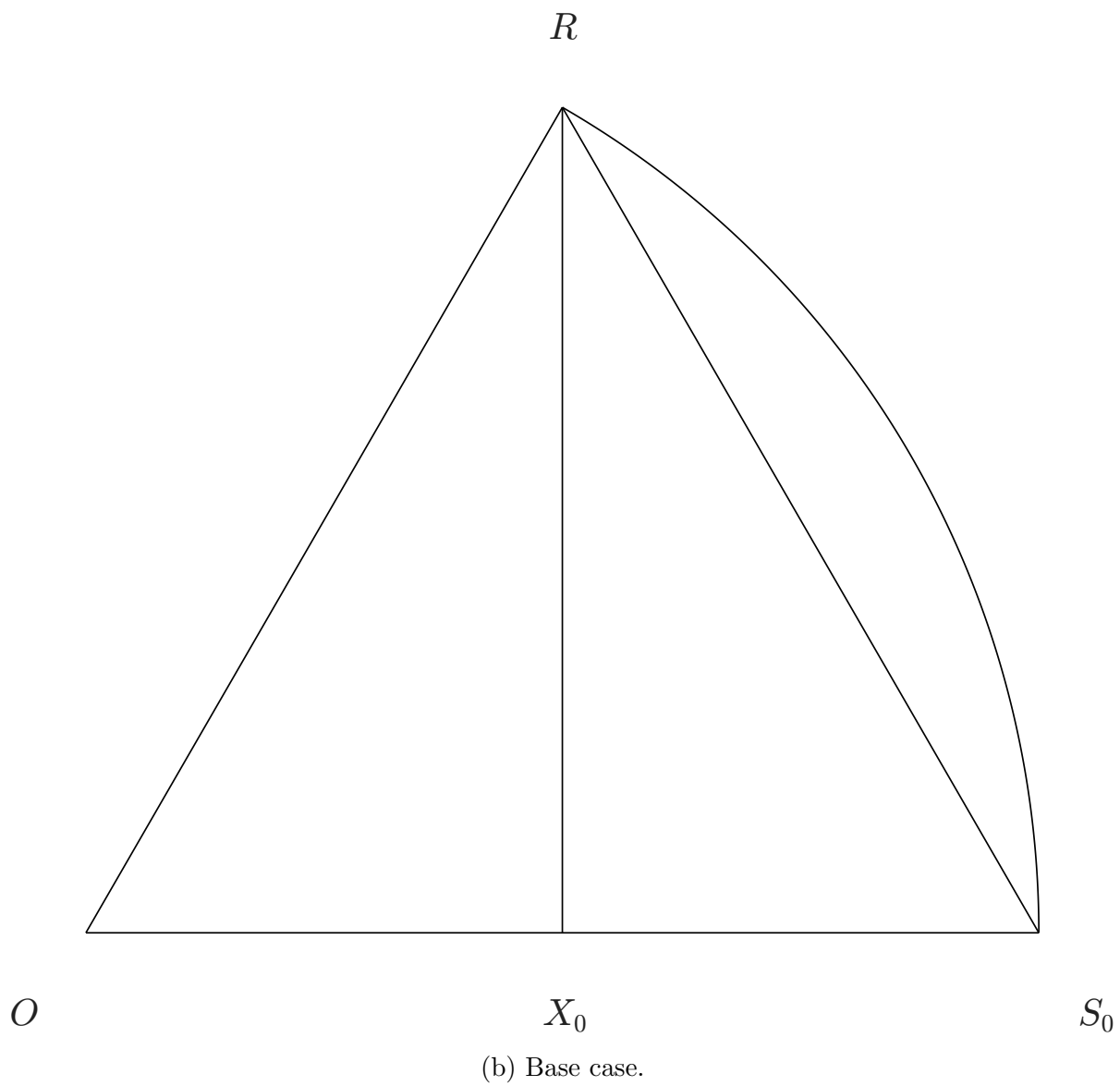


Figure 2: Circular sector sequence.

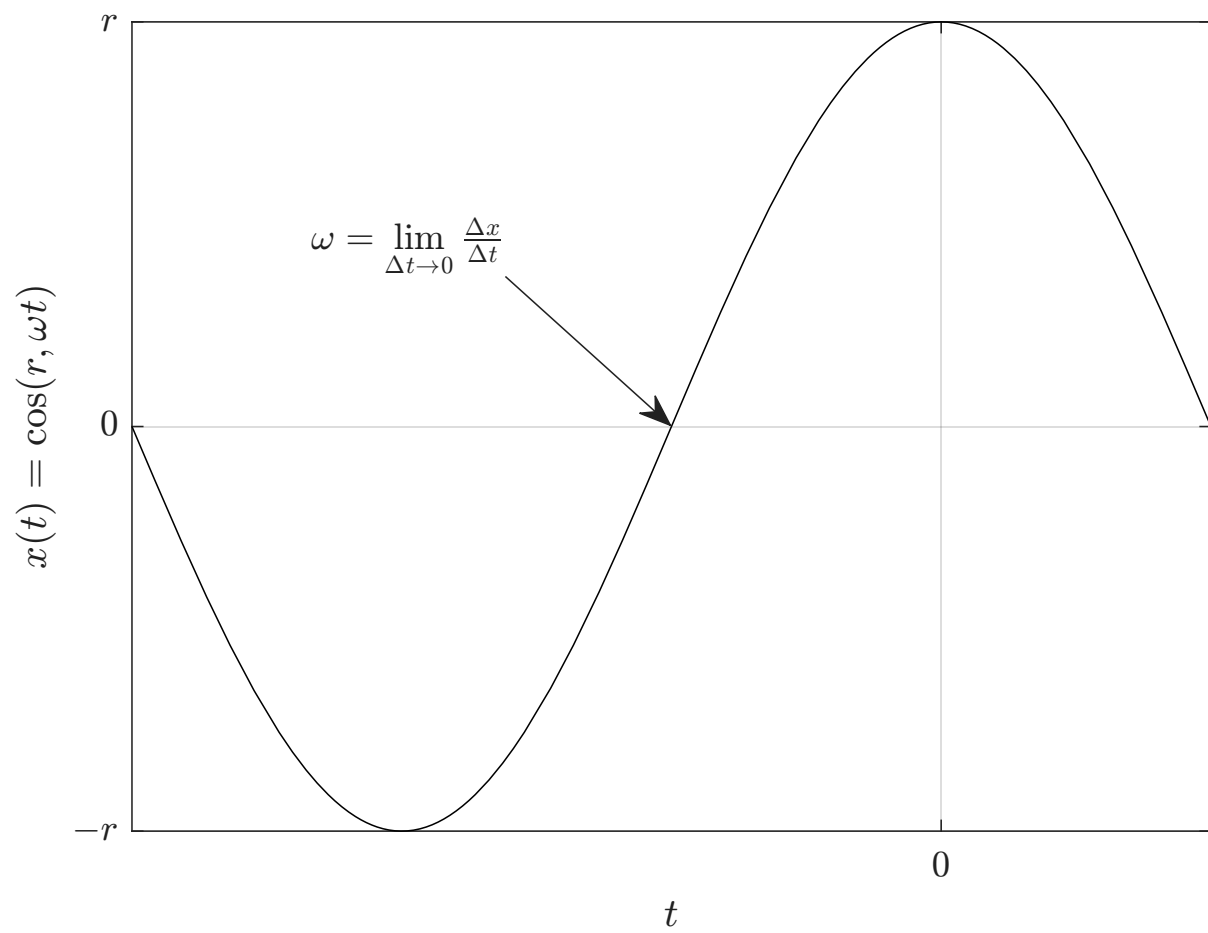


Figure 3: Simple harmonic motion.

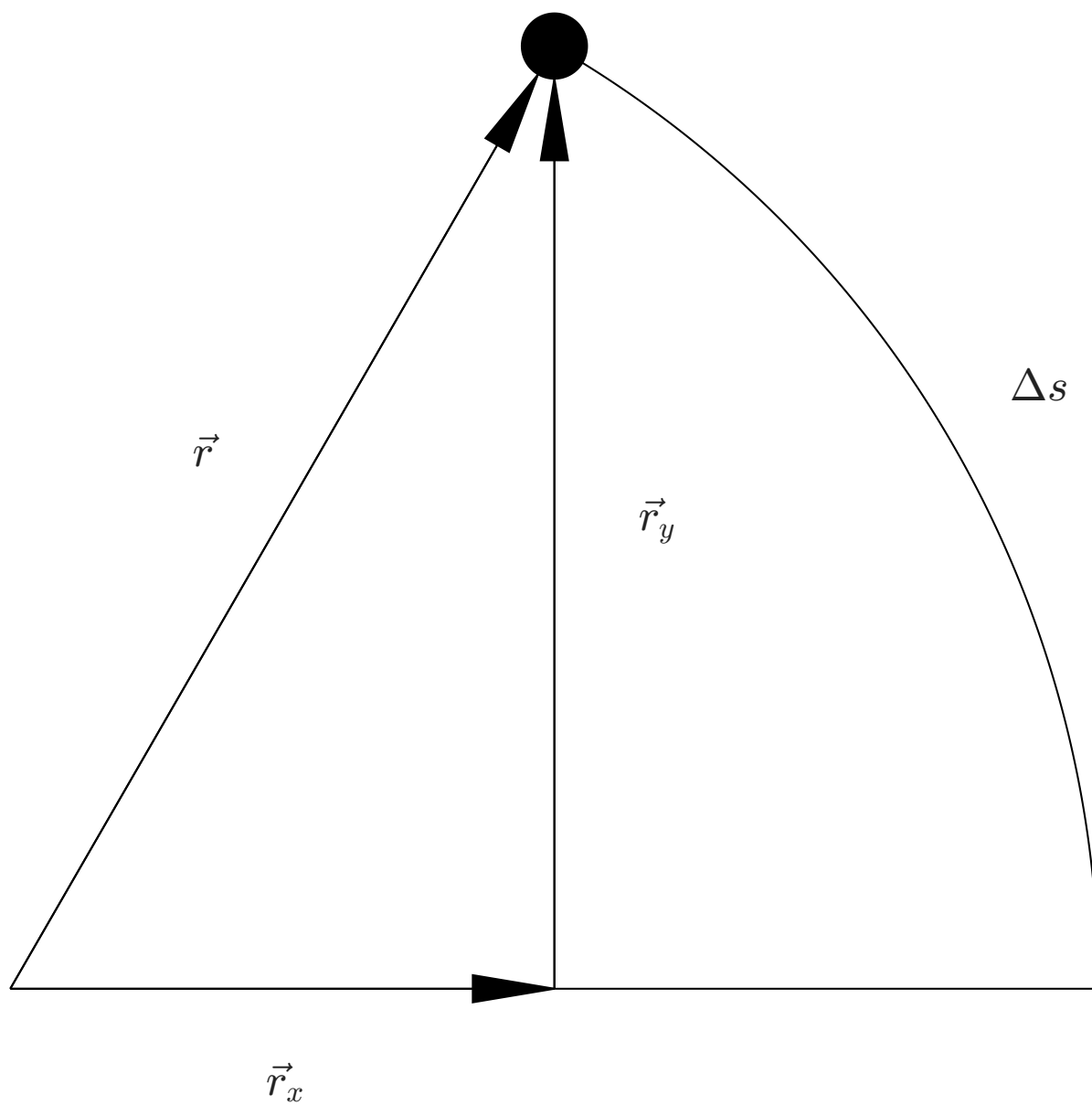


Figure 4: Uniform circular motion.